

The MINI series remote distributed IO modules support the EtherCAT protocol.



I. Product Overview

1.1 Product Introduction

The MINI series modules are compact in size and suitable for distributed wiring in large-scale equipment. This module can be placed directly near the actuator, and all IO signals can be pulled to it. The bus network ports are connected in a daisy chain manner to reduce the number of wires, improve production efficiency, and lower wiring costs.

The module itself is not waterproof. In harsh working conditions, a waterproof junction box should be added.

1.2 Product features

Small in size with a compact structure, it occupies little space, measuring only $123 \times 64 \times 44$ mm.

2. Rich variety of modules and complete range of I/O types

3. The Easy Assembly Wiring DIN 35mm standard rail installation adopts spring terminal blocks, making wiring convenient and fast.

4. Adopt EtherCAT dedicated communication protocol chips

Supports cascading up to 65,535 slave stations, with no restrictions on the topology structure.

The bus transmission rate is as high as 100 Mbps.

7. Provide XML files for simple configuration

1.3 Model list

Table 1.0

Model number	Module Description	Note
Digital quantity module		
XHS-ECT-ID16	16-channel digital input (PNP/NPN universal)	
XHS-ECT-OD16	16-channel digital output (NPN)	-P: PNP output
XHS-ECT-ID32	32-channel digital input (PNP/NPN universal)	General-purpose -P: PNP output
XHS-ECT-OD32	32-channel digital output (NPN)	
XHS-ECT-MD1616	16-channel digital input (PNP/NPN universal), 16-channel digital output (NPN)	
XHS-ECT-MD0808	8-channel digital input (PNP/NPN universal), 8-channel P: PNP output digital output (NPN)	
Analog module		

1.4 Naming rules

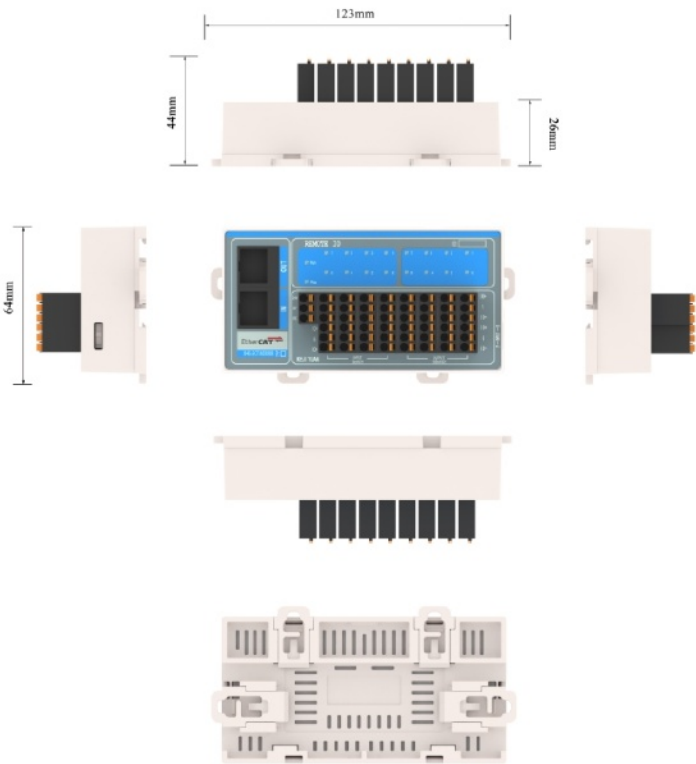
XHS - ECT - MD 1616 - P _

① ② ③ ④ ⑤

Table 1.1

series	② Bus category	③ Interface Type	④ Number of interfaces	⑤ Terminal polarity
XHS: Mini Type KJY: Blade type	ECT: EtherCAT MCP: MODBUSTCP EIP: EtherNetIP C C: CC LINKIE Basik P N: ProfiNET	MD: Input and Output ID: Input OD: Output	1616: 16 Input 0808: 8 Input 8 Output ID16: 16 Input ID32: 32 Input OD16:16 Output OD32: 32 outputs	P: PNP output 16 Output None: NPN Output

1.5 Specifications and dimensions



1.6 Terminal Block Description

Table 1.2

Numbering	Explanation	Note
IN (RJ45)	EtherCAT input interface	
OUT(RJ45)	EtherCAT output interface	
24V	Module power supply	
0V	Module power supply	
PE	Module grounding	Connecting to the ground wire can enhance the anti-interference ability.
IO+	Sensor power supply	No need, no connection.
IO-	Sensor power supply	No need to connect if not necessary.
0... 31	Input or output number	

1.7 Terminal layout

32-channel interface:

Figure 1: Power Distribution

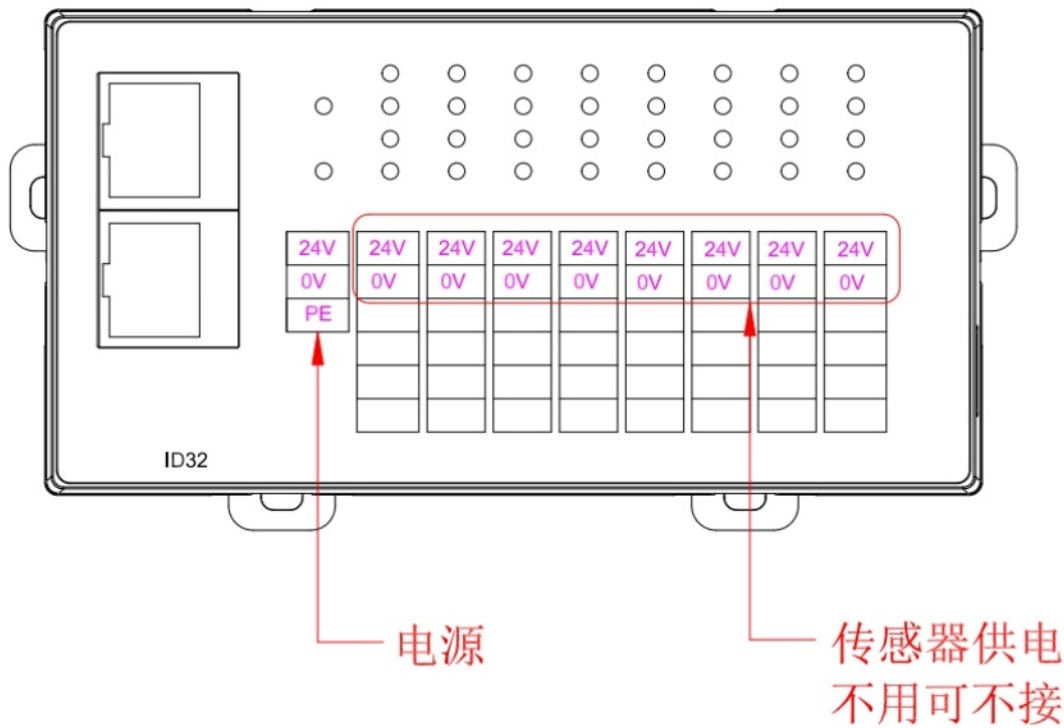


图 2:OD32 端子分布

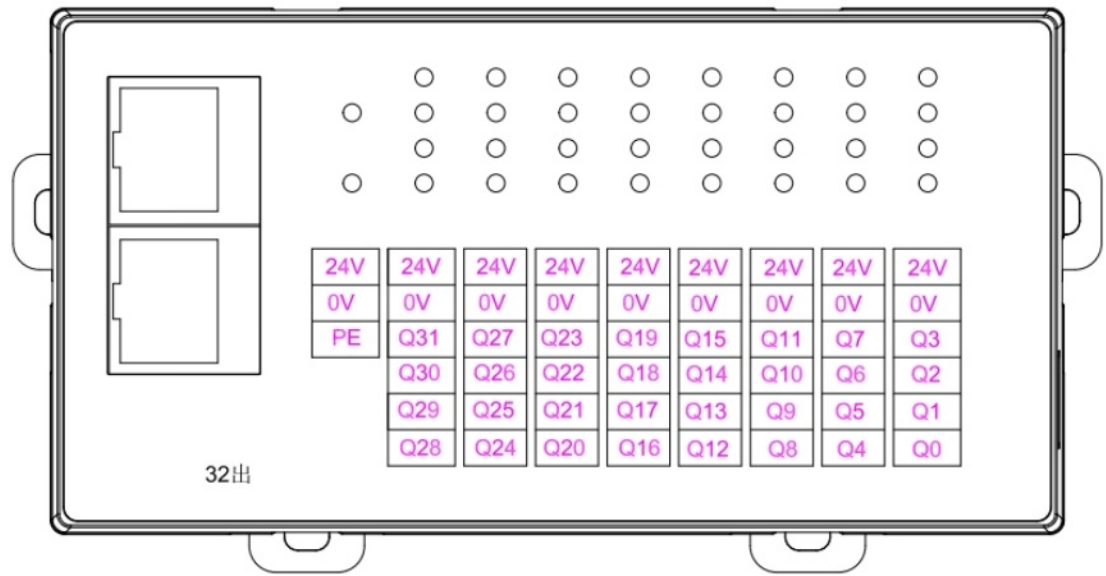


Figure 3: Terminal Distribution of ID32

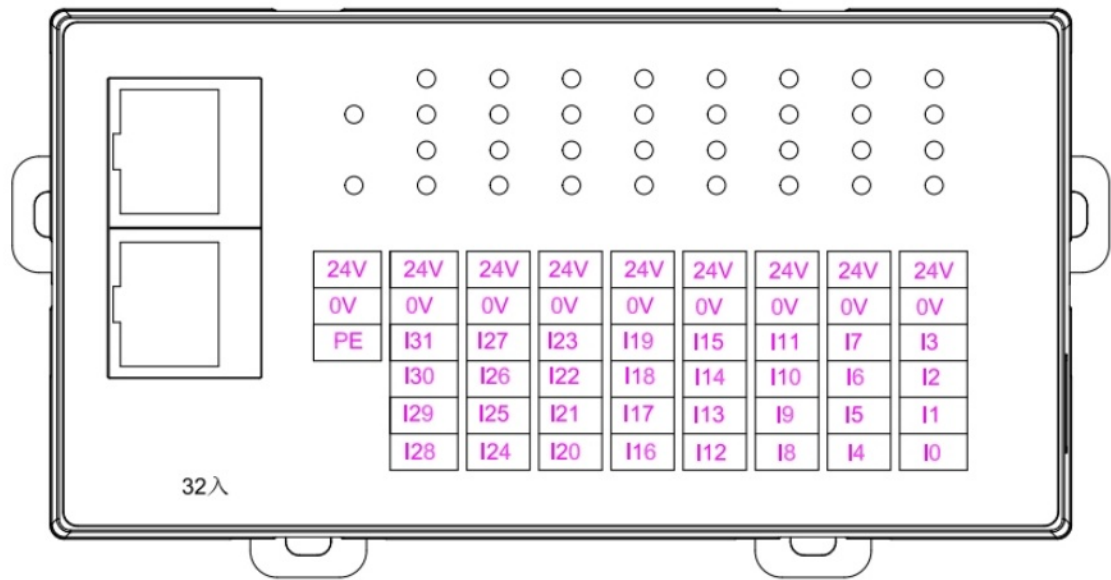
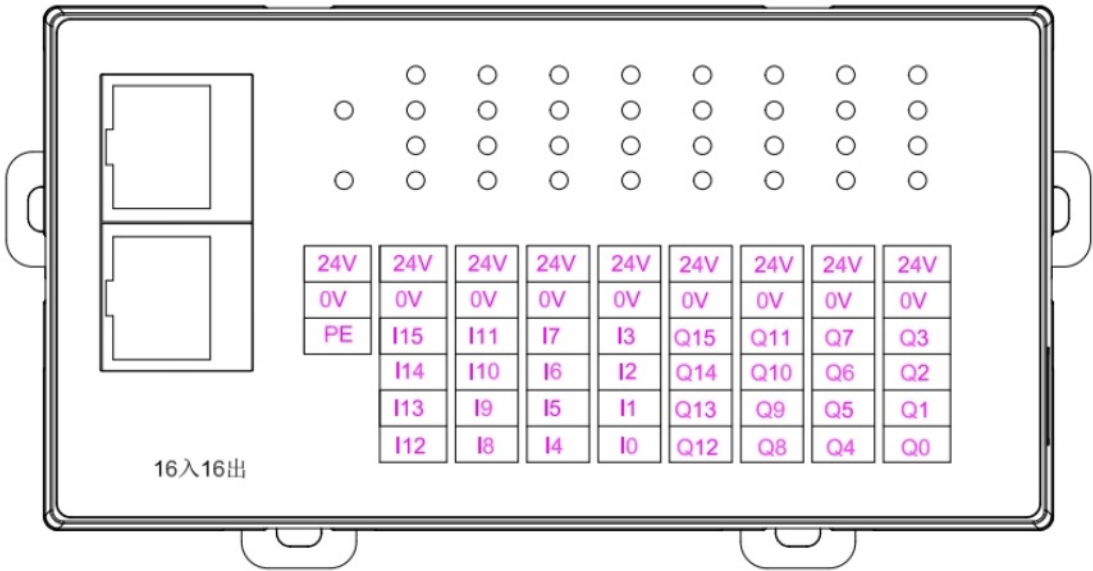


Figure 4: Terminal Distribution of MD1616



16-channel interface (each interface is equipped with power supply):

Figure 5: Power Distribution

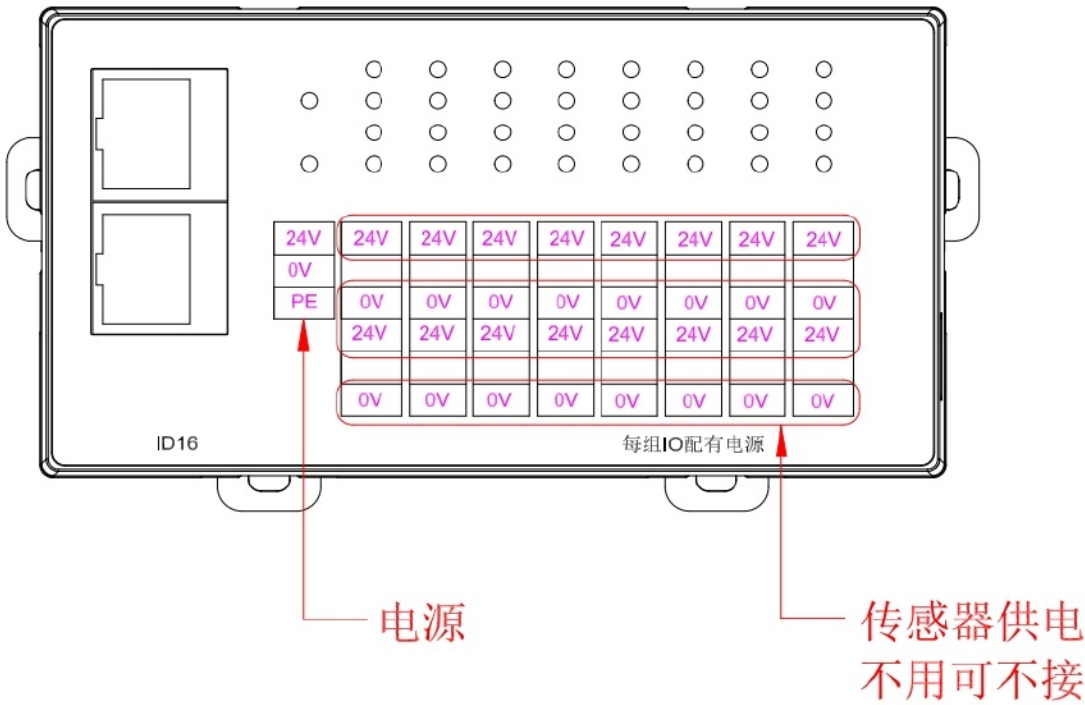


Figure 6: Terminal Distribution of OD16

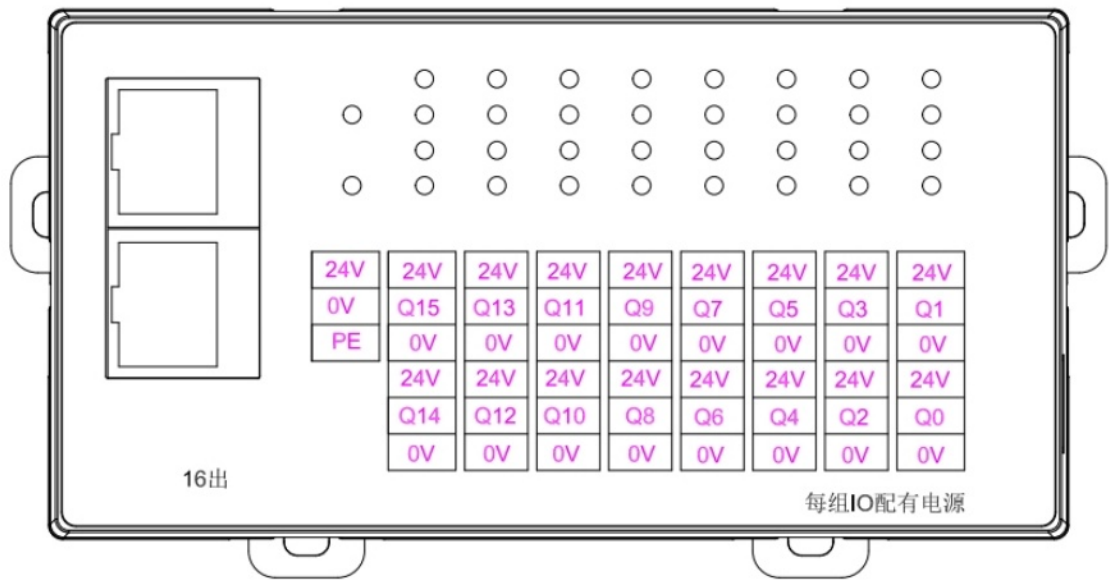


Figure 7: Terminal Distribution of ID16

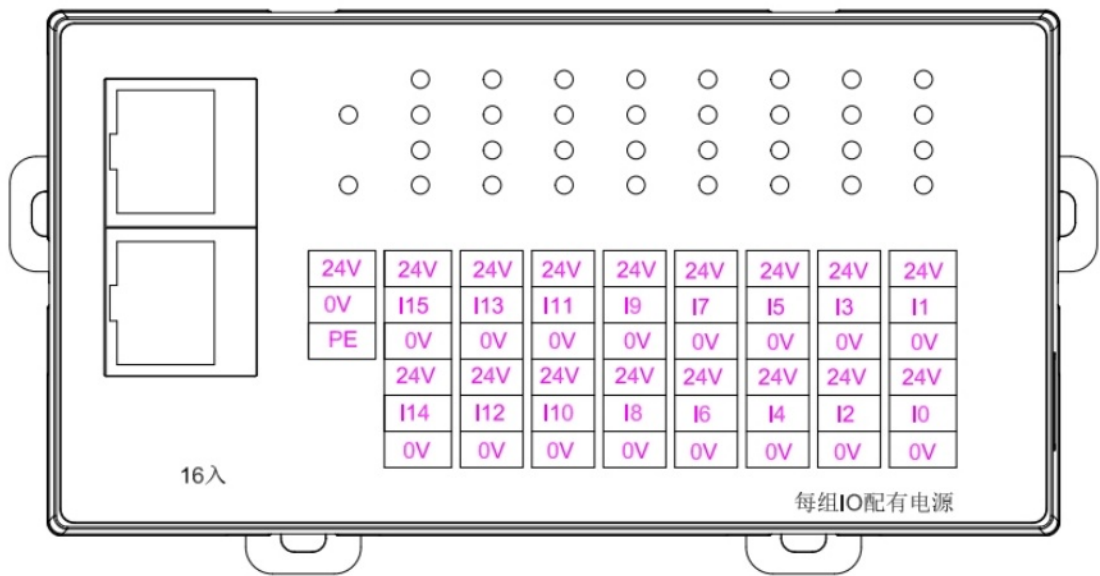
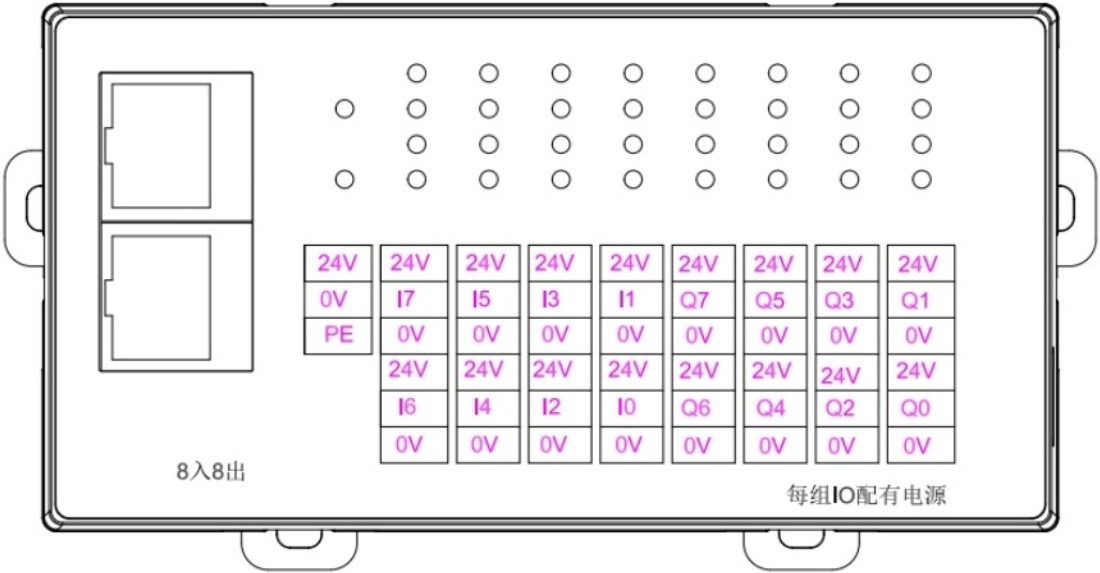


Figure 8: Terminal Distribution of MD0808



1.8 Wiring Example

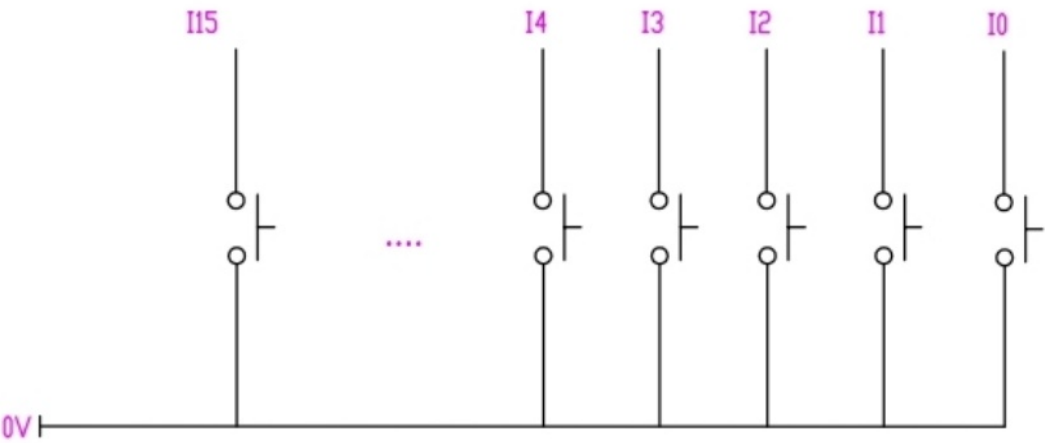


Figure 9: NPN Input Wiring Example

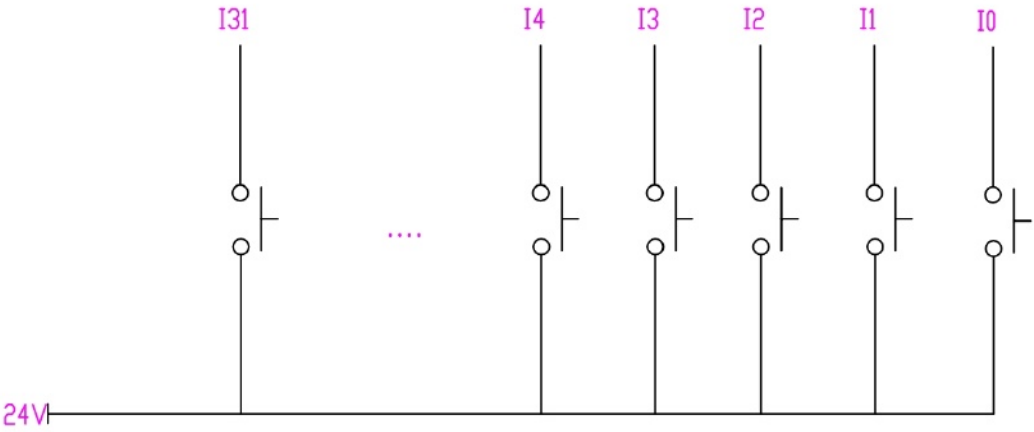


Figure 10: PNP Input Wiring Example

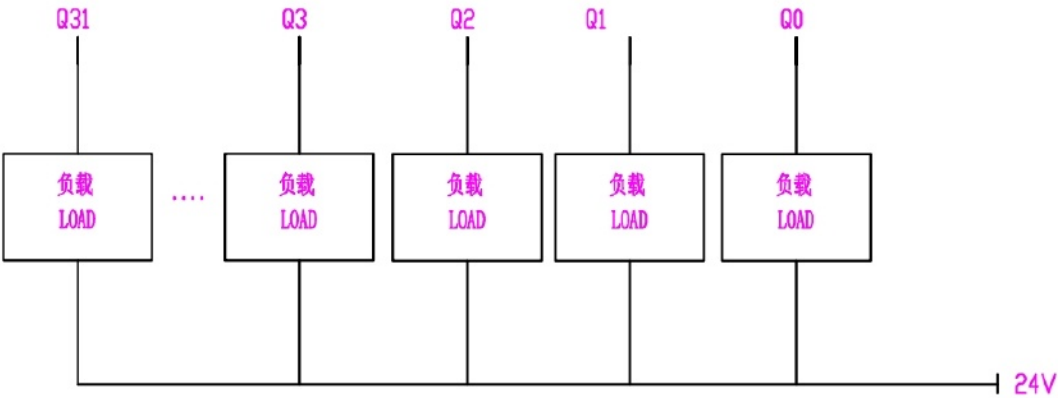


Figure 11: NPN Output Wiring Example

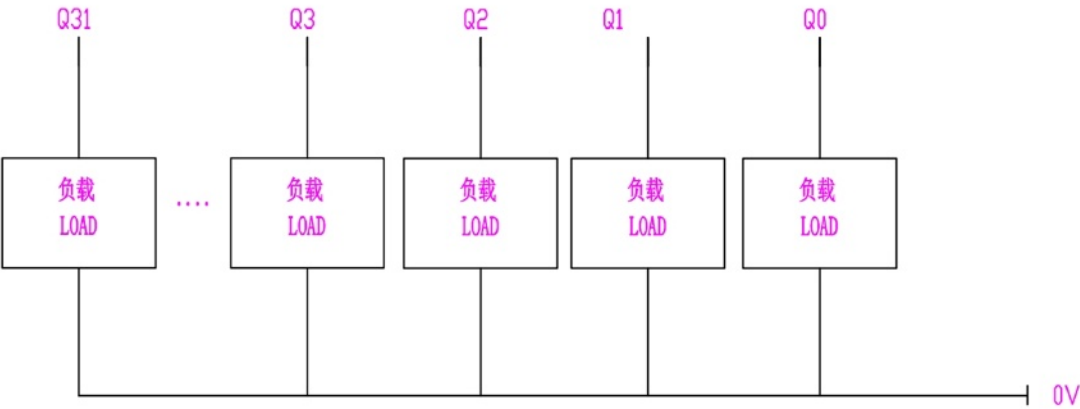


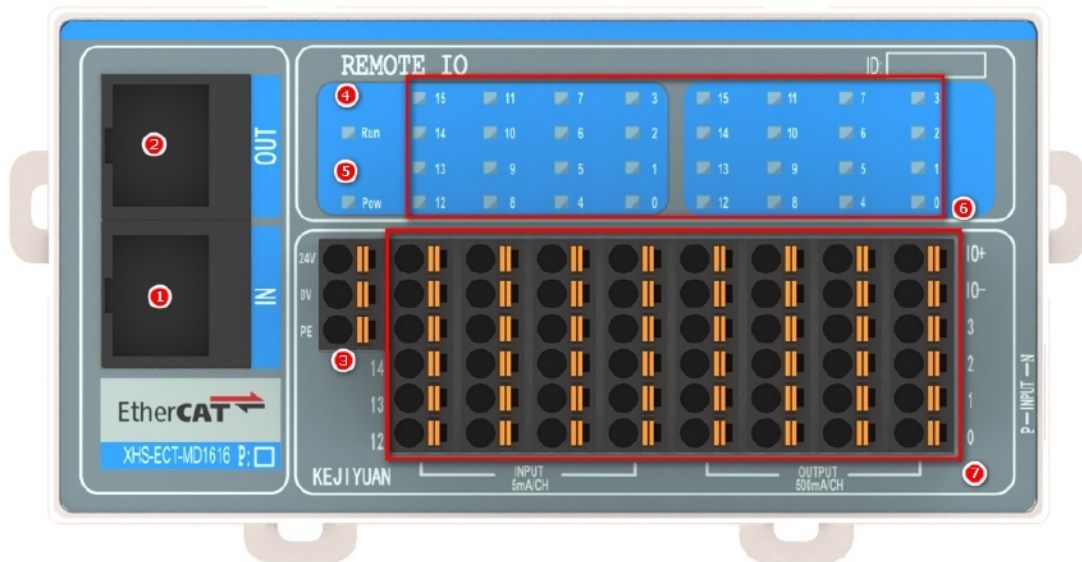
Figure 12: PNP Output Wiring Example

1.9 Equipment parameters

Table 1.3

Equipment parameters	
Power supply voltage	DC 12~30V
Merit expenditure	<200mW
Connection method	100M RJ45 network connection
Communication protocol	EtherCAT communication
IO input current / voltage	5mA per channel, DC 24V
IO output current / voltage	500mA per channel, DC24V. The total maximum current for every 8 channels does not exceed 2.5A.
Working temperature	-40°C to +85°C
Working humidity	0% to 75% RH (no condensation)
Installation method	Standard C45 (35mm) universal guide rail

1.10, Layout of the equipment panel



EtherCAT input interface

EtherCAT output interface

24V power supply

Operation status indicator light (steady: connected, flashing: not connected)

Power indicator light

Status indication of input and output interfaces

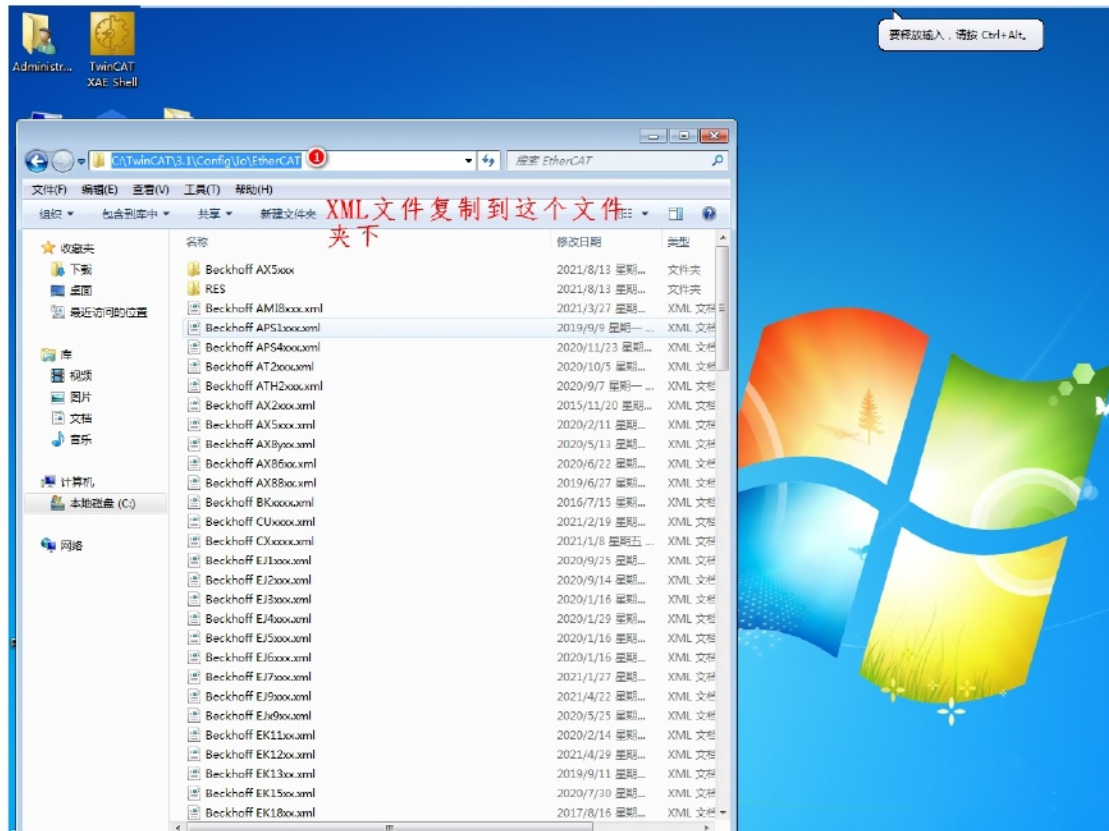
Input or output terminal blocks

II. Equipment Connection and Usage

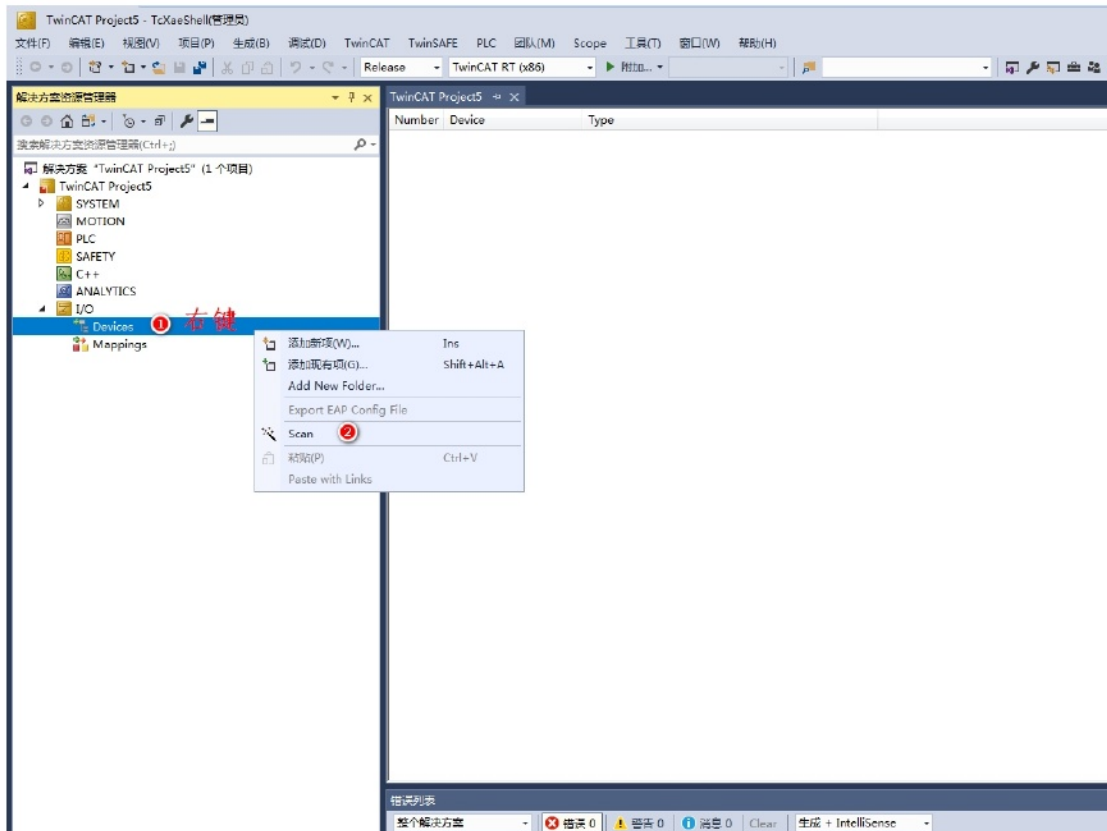
1.1 Equipment Compatibility Instructions with TwinCAT3

Copy the device XML configuration file to the installation path of TwinCAT3.

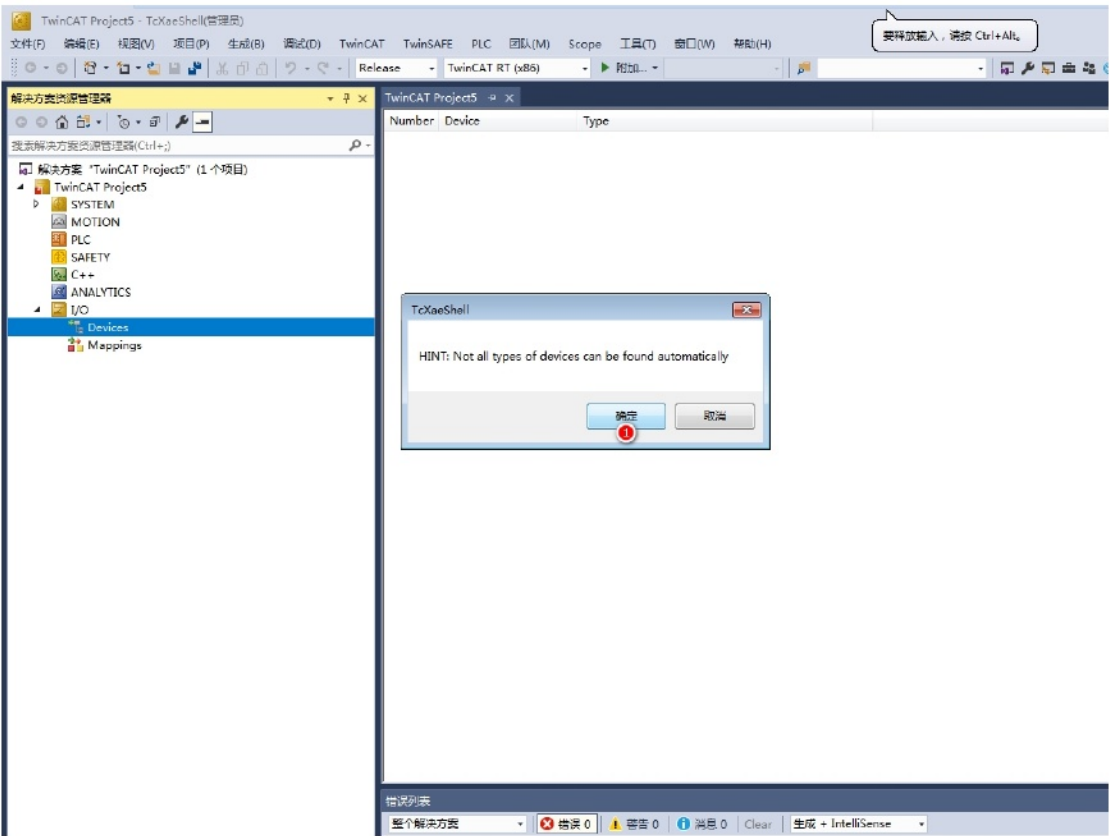
Such as: C:\TwinCAT3.1\ConfigVlo\EtherCAT



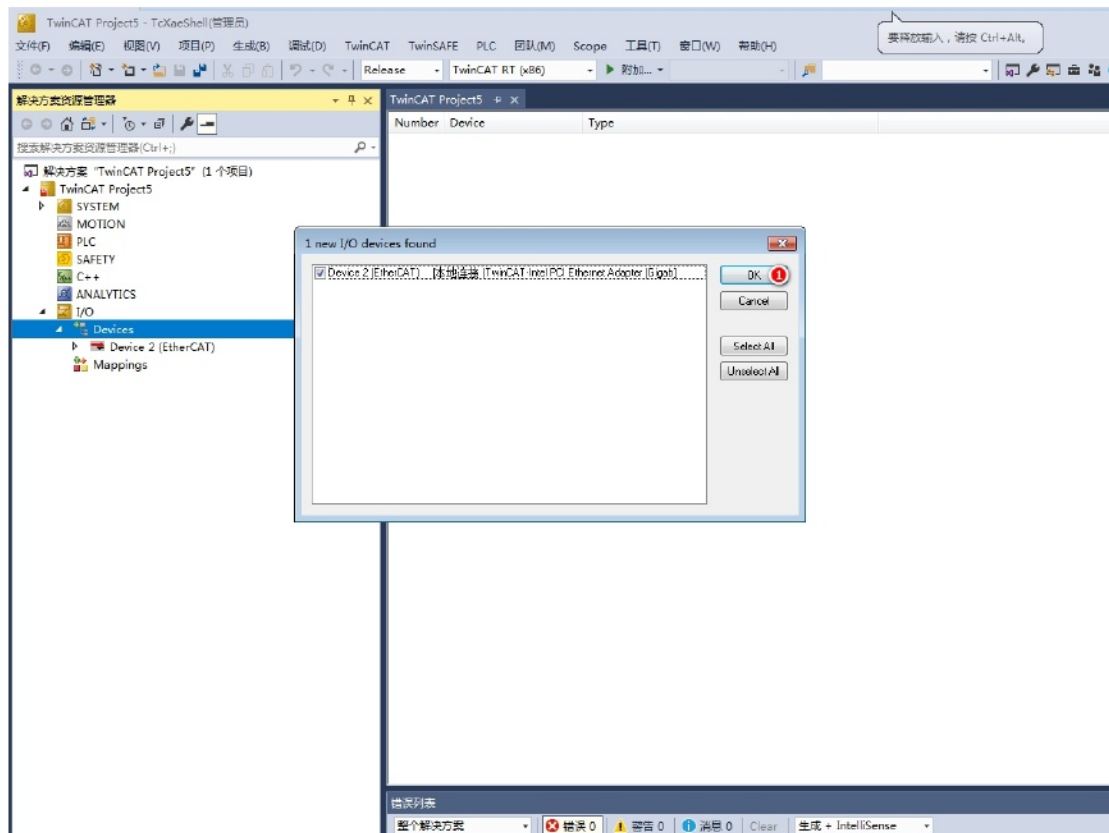
Open TwinCAT3 software and create a new project. Right-click on "Devices" in the project tree and then left-click on "Scan" to scan the devices.



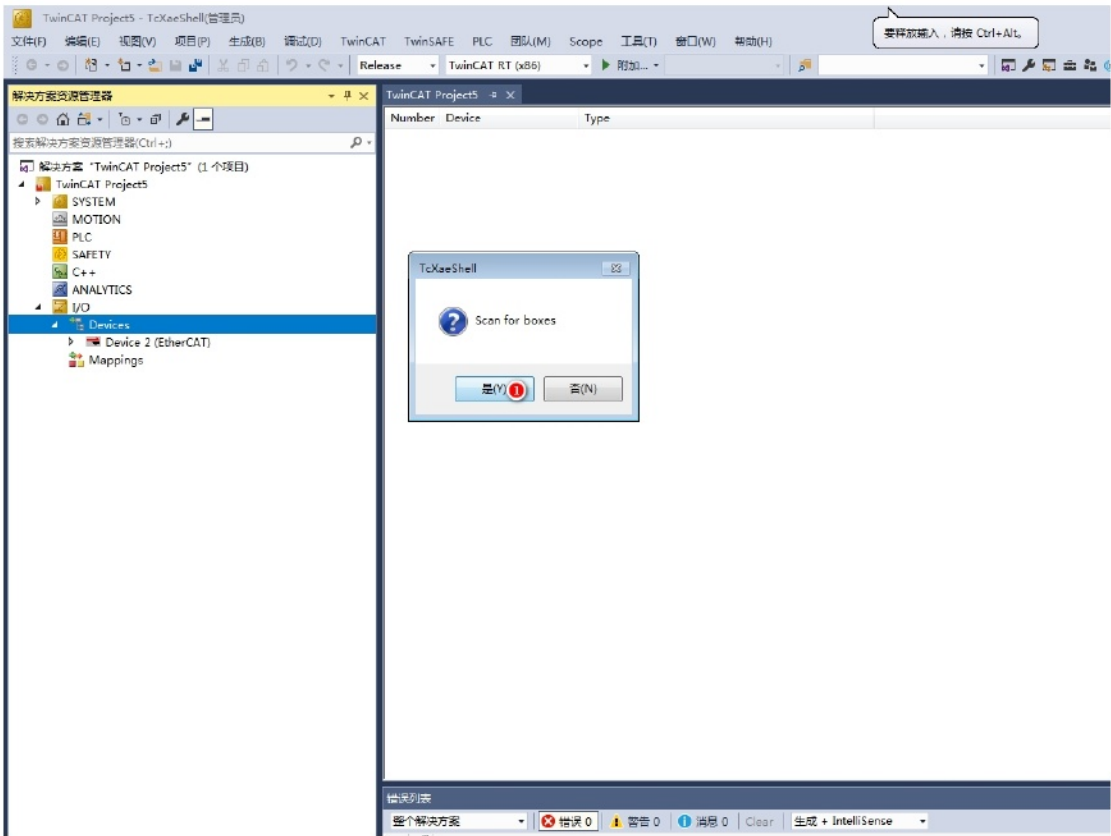
Click "OK" in the pop-up dialog box.



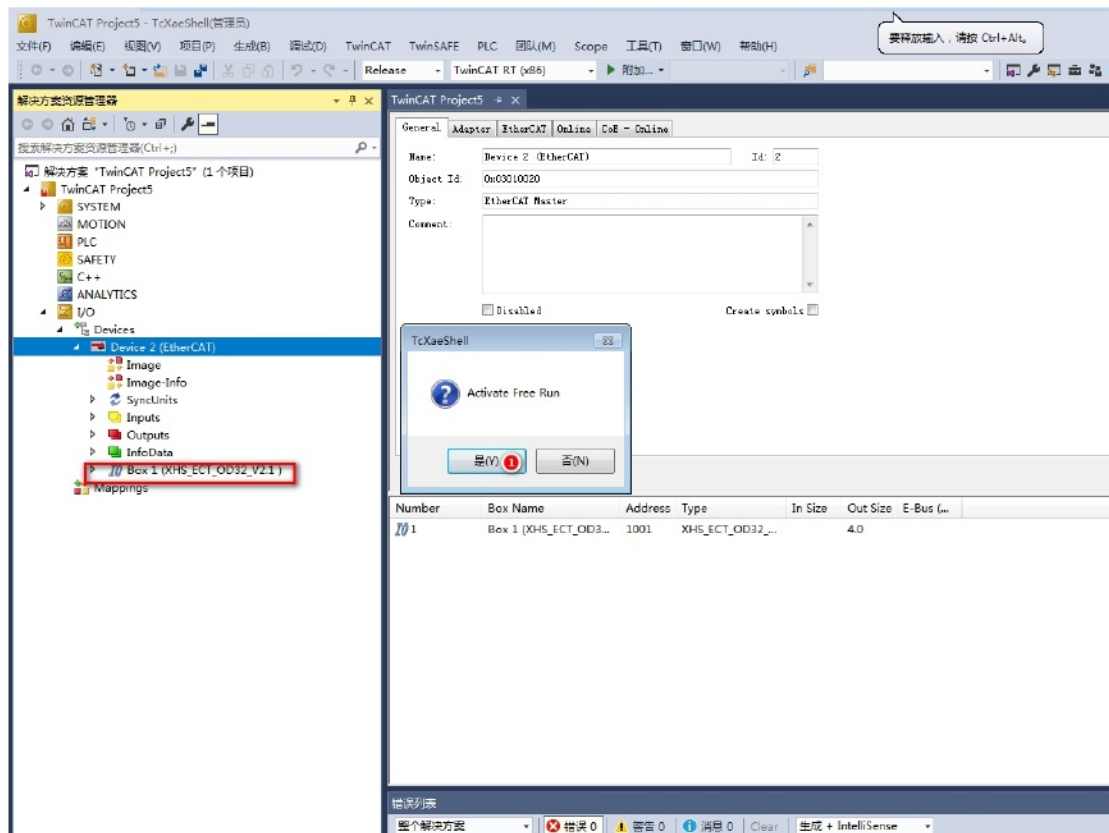
After connecting to the network interface, click "OK".



After scanning to the device, click "Yes".



After scanning to the device, it appears in the project tree and a dialog box pops up. Click "Yes" to activate and run.



Check whether the test equipment is working properly.

The model of the device in the sample diagram is: XHS-ECT-OD32. This module has 32 output channels, corresponding respectively to:

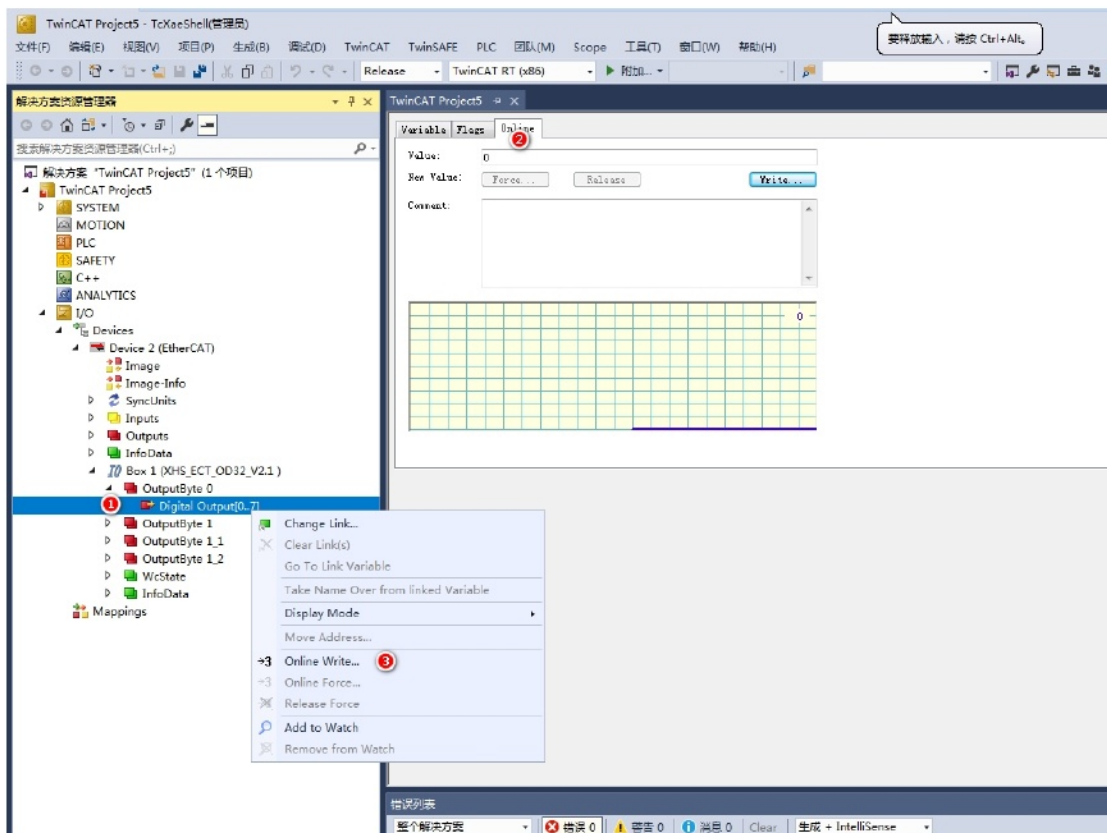
OutputByte 0,

OutputByte1,

OutputByte 2,

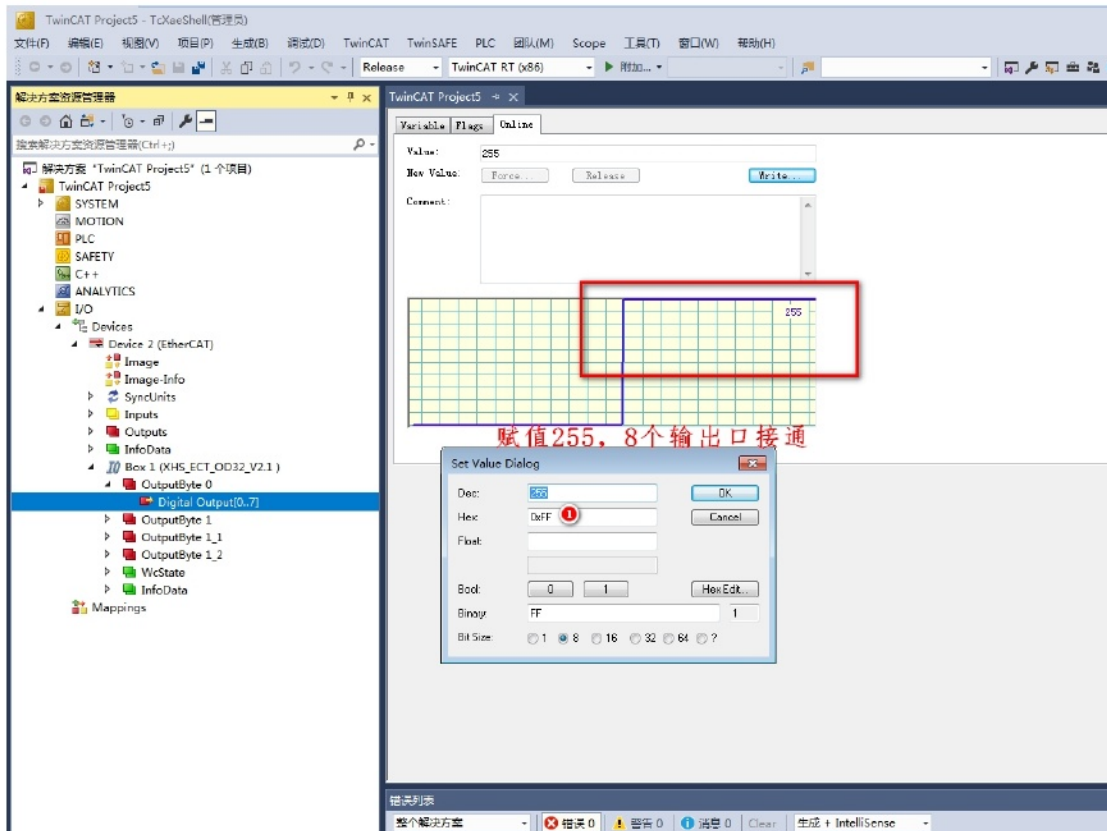
OutputByte 3,

Right-click on Digital Output [0..7] under OutputByte 0 and click "Online Write..." to enter the forced output dialog box.



Force assignment to 255, (all 8 output ports output)

At this time, the indicator light corresponding to the output port on the module lights up.



The interfaces of other groups can be tested in the same way.

III. Troubleshooting

Fault phenomenon	Elimination method
The power indicator light is not on.	Check whether the power interface 24V and 0V are normal.
The RUN indicator light is flashing.	The module is not connected to the network. Check if the network cable is properly connected. 2. Has the XML file been imported into the PLC programming software and downloaded to the PLC? And the PLC is in the running state. 3. Has the network cable from the superior device been connected to the "IN" interface of this device?
No signal at the input/output port.	Check the wiring according to the NPN or PNP wiring examples in this instruction.
Other faults	Please contact the equipment sales department.